

In [65]: `import pandas as pd`

In [66]: `data=pd.read_csv("C:\\Users\\Dell\\Desktop\\Work Data\\car_prices.csv")`

In [67]: `data.sample(10)`

Out[67]:

	year	make	model	trim	body	transmission	vin	state	condition	odometer	color	interior	seller	mmr	sellingprice	saledate
540041	2015	Mitsubishi	Mirage	ES	hatchback	automatic	ml32a4hj0fh032387	mo	49.0	486.0	white	black	santander consumer	9675.0	10300.0	Wed Jun 17 2015 03:30:00 GMT-0700 (PDT)
351948	2012	Ford	F-150	Platinum	SuperCrew	automatic	1ftfw1ef3cfa58746	mn	39.0	43725.0	red	black	ford motor credit company llc	32300.0	31900.0	Wed Feb 25 2015 02:00:00 GMT-0800 (PST)
432996	2007	NaN	NaN	NaN	NaN	automatic	salsk25417a112751	tx	36.0	111748.0	black	black	land rover capital/primus	16000.0	18500.0	Thu May 21 2015 03:30:00 GMT-0700 (PDT)
190068	2012	Honda	Odyssey	EX-L	Minivan	automatic	5fnrl5h63cb012413	pa	37.0	27452.0	gray	gray	ahfc/honda lease trust/hvt inc. eot	23400.0	21100.0	Fri Jan 30 2015 01:00:00 GMT-0800 (PST)
279075	2011	Toyota	Corolla	LE	Sedan	automatic	2t1bu4ee6bc557237	tn	2.0	47125.0	gray	gray	tdaf remarketing	9800.0	10500.0	Wed Feb 11 2015 02:30:00 GMT-0800 (PST)
193016	2012	Mercedes-Benz	C-Class	C300 Luxury 4MATIC	Sedan	automatic	wddgfg8bb1cr200224	wi	37.0	32849.0	black	black	mercedes-benz financial services	21200.0	18000.0	Wed Jan 28 2015 02:00:00 GMT-0800 (PST)
392594	2003	Ford	F-150	SVT Lightning	Regular Cab	automatic	2ftrf073x3cb00271	nc	38.0	154675.0	white	gray	pro cycle	7625.0	9300.0	Tue Mar 03 2015 02:00:00 GMT-0800 (PST)
285820	2011	Dodge	Grand Caravan	Crew	Minivan	automatic	2d4rn5dg7br690175	mn	39.0	162827.0	charcoal	black	ge fleet services for itself/servicer	6525.0	6700.0	Thu Feb 12 2015 02:45:00 GMT-0800 (PST)
248435	2014	GMC	Savana Cargo	2500	Van	automatic	1gtw7fca5e1904215	hi	36.0	4683.0	white	gray	u-haul	22500.0	21000.0	Wed Feb 11 2015 06:35:00 GMT-0800 (PST)
331637	2004	Toyota	Camry	LE	Sedan	automatic	4t1be32k64u339396	mo	24.0	136087.0	black	gray	plaza motor company	4150.0	3900.0	Tue Feb 17 2015 03:00:00 GMT-0800 (PST)

In [68]: `data.isna().sum()`

Out[68]:

year0

make10301

model10399

trim10651

body13195

transmission65352

vin4

state0

condition11820

odometer94

color749

interior749

seller0

mmr38

sellingprice12

saledate12

dtype: int64

In [69]: `for i in data.columns:
 data.rename(columns={i:i.title()}, inplace=True)
data.columns`

```
Out[69]: Index(['Year', 'Make', 'Model', 'Trim', 'Body', 'Transmission', 'Vin', 'State',
        'Condition', 'Odometer', 'Color', 'Interior', 'Seller', 'Mmr',
        'Sellingprice', 'Saledate'],
        dtype='object')

In [70]: data.head()

Out[70]:
```

	Year	Make	Model	Trim	Body	Transmission	Vin	State	Condition	Odometer	Color	Interior	Seller	Mmr	Sellingprice	Saledate
0	2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg566472	ca	5.0	16639.0	white	black	kia motors america inc	20500.0	21500.0	Tue Dec 16 2014 12:30:00 GMT-0800 (PST)
1	2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg561319	ca	5.0	9393.0	white	beige	kia motors america inc	20800.0	21500.0	Tue Dec 16 2014 12:30:00 GMT-0800 (PST)
2	2014	BMW	3 Series	328i SULEV	Sedan	automatic	wba3c1c51ek116351	ca	45.0	1331.0	gray	black	financial services remarketing (lease)	31900.0	30000.0	Thu Jan 15 2015 04:30:00 GMT-0800 (PST)
3	2015	Volvo	S60	T5	Sedan	automatic	yv1612tb4f1310987	ca	41.0	14282.0	white	black	volvo na rep/world omni	27500.0	27750.0	Thu Jan 29 2015 04:30:00 GMT-0800 (PST)
4	2014	BMW	6 Series Gran Coupe	650i	Sedan	automatic	wba6b2c57ed129731	ca	43.0	2641.0	gray	black	financial services remarketing (lease)	66000.0	67000.0	Thu Dec 18 2014 12:30:00 GMT-0800 (PST)

```
In [71]: data.dtypes

Out[71]: Year          int64
        Make          object
        Model          object
        Trim           object
        Body           object
        Transmission    object
        Vin            object
        State          object
        Condition       float64
        Odometer        float64
        Color           object
        Interior        object
        Seller          object
        Mmr            float64
        Sellingprice    float64
        Saledate        object
        dtype: object

In [72]: data["Transmission"].describe()

Out[72]: count          493485
        unique             4
        top      automatic
        freq          475915
        Name: Transmission, dtype: object

In [73]: data1=data.groupby("Transmission")["Sellingprice"].sum()

In [74]: data1

Out[74]: Transmission
        Sedan      1.505500e+05
        automatic  6.443934e+09
        manual     1.966936e+08
        sedan      2.047500e+05
        Name: Sellingprice, dtype: float64

In [75]: filing=["Transmission", "Make", "Model", "Trim", "Body", "Color", "Interior"]
        data[filing]=data[filing].fillna("Unknown")

In [76]: data.isna().sum()

Out[76]: Year          0
        Make          0
        Model          0
        Trim           0
        Body           0
        Transmission    0
        Vin            4
        State          0
        Condition     11820
        Odometer        94
        Color           0
        Interior        0
        Seller          0
        Mmr            38
        Sellingprice    12
        Saledate        12
        dtype: int64

In [77]: a=data["Condition"].median()
        b=data["Condition"].mean()
        print(a)
        print(b)

35.0
30.672364844237016

In [78]: nfiling=["Condition", "Odometer", "Mmr", "Sellingprice"]
        data[nfiling]=data[nfiling].fillna(data[nfiling].median())
```

```
In [79]: data.isna().sum()

Out[79]: Year          0
        Make          0
        Model         0
        Trim          0
        Body          0
        Transmission  0
        Vin           4
        State         0
        Condition     0
        Odometer      0
        Color         0
        Interior      0
        Seller        0
        Mmr           0
        Sellingprice  0
        Saledate      12
        dtype: int64

In [80]: data=data.dropna(subset=["Vin", "Saledate"])

In [81]: data.isna().sum()

Out[81]: Year          0
        Make          0
        Model         0
        Trim          0
        Body          0
        Transmission  0
        Vin           0
        State         0
        Condition     0
        Odometer      0
        Color         0
        Interior      0
        Seller        0
        Mmr           0
        Sellingprice  0
        Saledate      0
        dtype: int64

In [82]: data["Saledate"]=pd.to_datetime(data["Saledate"],utc=True, errors='coerce')

C:\Users\Dell\AppData\Local\Temp\ipykernel_9712\7843088896.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`.
To ensure parsing is consistent and as-expected, please specify a format.
  data["Saledate"]=pd.to_datetime(data["Saledate"],utc=True, errors='coerce')

In [83]: data["Odometer"]=data["Odometer"].astype(int)

In [84]: data.dtypes

Out[84]: Year          int64
        Make          object
        Model         object
        Trim          object
        Body          object
        Transmission  object
        Vin           object
        State         object
        Condition     float64
        Odometer      int64
        Color         object
        Interior      object
        Seller        object
        Mmr           float64
        Sellingprice  float64
        Saledate      datetime64[ns, UTC]
        dtype: object

In [85]: upper_convert=["Vin", "State"]
        for i in upper_convert:
            data[i]=data[i].astype(str).str.upper()

In [86]: title_convert=["Make", "Model", "Trim", "Body", "Seller", "Color", "Interior"]
        for i in title_convert:
            data[i]=data[i].astype(str).str.title()

In [88]: lower_convert=["Transmission"]
        for i in lower_convert:
            data[i]=data[i].astype(str).str.lower()

In [ ]:

In [89]: data.head(10)
```

Out[89]:	Year	Make	Model	Trim	Body	Transmission	Vin	State	Condition	Odometer	Color	Interior	Seller	Mmr	Sellingprice	Saledate
0	2015	Kia	Sorento	Lx	Suv	automatic	5XYKTCa69FG566472	CA	5.0	16639	White	Black	Kia Motors America Inc	20500.0	21500.0	2014-12-16 04:30:00+00:00
1	2015	Kia	Sorento	Lx	Suv	automatic	5XYKTCa69FG561319	CA	5.0	9393	White	Beige	Kia Motors America Inc	20800.0	21500.0	2014-12-16 04:30:00+00:00
2	2014	Bmw	3 Series	328i Sulev	Sedan	automatic	WBA3C1C51EK116351	CA	45.0	1331	Gray	Black	Financial Services Remarketing (Lease)	31900.0	30000.0	2015-01-14 20:30:00+00:00
3	2015	Volvo	S60	T5	Sedan	automatic	YV1612TB4F1310987	CA	41.0	14282	White	Black	Volvo Na Rep/World Omni	27500.0	27750.0	2015-01-28 20:30:00+00:00
4	2014	Bmw	6 Series Gran Coupe	650i	Sedan	automatic	WBA6B2C57ED129731	CA	43.0	2641	Gray	Black	Financial Services Remarketing (Lease)	66000.0	67000.0	2014-12-18 04:30:00+00:00
5	2015	Nissan	Altima	2.5 S	Sedan	automatic	1N4AL3AP1FN326013	CA	1.0	5554	Gray	Black	Enterprise Vehicle Exchange / Tra / Rental / T...	15350.0	10900.0	2014-12-30 04:00:00+00:00
6	2014	Bmw	M5	Base	Sedan	automatic	WBSFV9C51ED593089	CA	34.0	14943	Black	Black	The Hertz Corporation	69000.0	65000.0	2014-12-17 04:30:00+00:00
7	2014	Chevrolet	Cruze	1Lt	Sedan	automatic	1G1PC5S82E7128460	CA	2.0	28617	Black	Black	Enterprise Vehicle Exchange / Tra / Rental / T...	11900.0	9800.0	2014-12-16 05:00:00+00:00
8	2014	Audi	A4	2.0T Premium Plus Quattro	Sedan	automatic	WAUFFAFL3EN030343	CA	42.0	9557	White	Black	Audi Mission Viejo	32100.0	32250.0	2014-12-18 04:00:00+00:00
9	2014	Chevrolet	Camaro	Lt	Convertible	automatic	2G1FB3D37E9218789	CA	3.0	4809	Red	Black	D/M Auto Sales Inc	26300.0	17500.0	2015-01-19 20:00:00+00:00

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In [90]: data["State"]=data["State"].apply(lambda x: "Unknown" if len(str(x))>5 else x)

In [91]: data.to_csv("C:\\Users\\Dell\\Desktop\\Work Data\\cleaned_car_prices.csv", index=False)

In [ ]:
```